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Prof. Dr. Alexander Moradi

The Impact of Home Allergen Reduction Kits on Health Outcomes in Individuals with Autoimmune Skin Disorders

Enio Hoxha

Atopic dermatitis (AD) is a chronic inflammatory skin condition characterized by recurrent flare ups, intense itching, and substantial impairment in daily functioning. In Italy, AD affects up to 8% of the adult population (Pocetta et al., 2014), generating a considerable clinical and economic burden through increased healthcare utilization, reduced productivity, and impaired sleep and wellbeing.

Several clinical studies document the role of indoor environmental factors in triggering AD symptoms. In particular, dust mite allergens are consistently associated with disease exacerbations, flare ups, and persistent inflammation (Huang et al., 2023). Broadly, exposure to airborne pollutants, volatile chemicals, and household irritants has been linked to increased flare frequency, greater symptom severity (Silverberg, 2017), and sleep disruption in pediatric populations (Chang et al., 2016), with similar mechanisms plausibly affecting adults. Despite this evidence, public health responses to AD remain predominantly pharmacological, with limited emphasis on preventive, home based environmental interventions. Adoption of effective allergen reduction tools such as anti-mite mattress encasings, HEPA air purifiers, and low-irritant detergents, remains low, particularly among lower income households. Barriers include upfront costs and limited awareness, even though AD is associated with substantial healthcare expenditures and productivity losses (Drucker et al., 2017; Chung & Simpson, 2019). This difference between clinical evidence and everyday adoption suggests the presence of informational and financial constraints, resulting in underinvestment in preventive measures with potentially high social returns. This study addresses this policy gap by evaluating whether the provision of a standardized allergen reduction kit can reduce flare frequency and improve patient outcomes among adults with AD. Using a rigorously implemented randomized controlled trial, the project aims to generate credible causal evidence on the effectiveness of environmental interventions as a complement to standard AD treatment, informing the design and potential scale up of preventive public health policies.

The proposed intervention is a standardized, nonmedical, allergen reduction kit. The kit includes impermeable anti-mite mattress and pillow encasings, which create a physical barrier preventing contact with mite allergens accumulating in bedding. Given that individuals spend nearly one-third of the day in bed, this component directly targets one of the most constant and concentrated exposure points. Second component is a portable HEPA air purifier, Technology capable of filtering up to 99.97% of air particles, thus reducing indoor particulate matter such as dust, pollen, molds, and volatile irritants. Environmental irritants have been associated with increased flare frequency, impaired sleep, and more severe symptoms (Silverberg, 2017). HEPA filtration therefore aims to improve the nighttime microenvironment and reduce airway and skin irritation. Finally, the kit includes a fragrance free and low irritant detergent. Many detergents contain surfactants and fragrances that damage the skin barrier or provoke irritant reactions, by standardizing detergents reduces heterogeneity in chemical exposures across households. The total cost of the kit is 260€ per kit (Table 1). The kit's components are united by a clear mechanistic rationale: reducing environmental triggers that influence flare cycles. Unlike pharmacological treatment which targets inflammation downstream, this intervention aims at reducing upstream exposures that sustain chronic symptoms.

A key identification challenge concern is the definition of the eligible population, which must simultaneously ensure internal validity and policy relevance. Key to this topic is that many components of the allergen reduction kit such as anti-mite encasings, HEPA purifiers, and low-irritant detergents may already be used by some households. Including such participants would attenuate the treatment contrast, induce downward biased estimates and potentially obscure the true causal effect. Conceptually this is analogous to evaluating the impact of distributing toothbrushes to individuals who already brush their teeth daily: the intervention would not meaningfully change behavior or

exposure. To prevent this, the study introduces a large pre-screening survey (N ≈ 1,000), which collects information on AD diagnosis, flare frequency, housing conditions, and crucially baseline use of allergen control tools. Eligibility for the randomized controlled trial is restricted to individuals reporting no regular use of mattress and pillow encasings, no ownership of HEPA purifiers, and no reliance on low irritant detergents become eligible for the RCT. The resulting target population consists of adults with clinically diagnosed AD, and lacking access to protective environmental tools This design choice is directly aligned with policy objectives, as the targets households facing financial or informational barriers to adoption. By focusing on non-users with documented exposure risks, the evaluation estimates the causal effect for the population most likely to benefit from environmental improvements, thereby enhancing both the internal validity and external policy relevance of the findings.

Following with the experimental design, the study adopts an individually randomized controlled trial (RCT) with a 1:1 allocation ratio between treatment and control groups. After completion of the pre-screening survey and confirmation of eligibility, participants are assigned to groups through randomization, ensuring that treatment assignment is fully exogenous. To improve precision and reduce variance in estimated treatment effects, the randomization is stratified along pre-specified variables: (i) baseline AD severity (mild vs. moderate), (ii) environmental exposure category (high vs. low humidity/ventilation). Stratification guarantees balance across groups on key predictors of outcomes, strengthening internal validity without introducing bias since it is performed prior to treatment assignment. Outcome assessors performing clinical evaluations remain blinded to treatment status, limiting observer bias. To reduce differential reporting, participants are informed that the study evaluates environmental conditions but are not explicitly told which outcomes are expected to change. Data are collected at three time points: baseline, a midline follow-up at six months, and an endline assessment at twelve months. Key performance indicators are clinical measures, including Eczema Area and Severity Index (EASI) and self-reported flare diaries. This design directly controls for confounding variables through randomization, mitigates measurement bias through blinding and validated instruments, and allows for credible estimation of causal effects on both short-term and longer-term AD outcomes.

The Econometric Strategy is as follows, the primary estimand of interest is the Intent to Treat (ITT) effect, defined as the average causal effect of being offered the allergen reduction kit. The ITT is policy relevant because policymakers can decide whether to offer the intervention but cannot directly control individual uptake. The ITT effect is estimated using the following linear model:

$$Y_i = \alpha + \beta T_i + \gamma' X_i + \varepsilon_i,$$

where $Y_i$denotes the outcome (monthly flare count, EASI score), T_i is random assignment to the treatment arm, and $X_i$represents pre-registered baseline covariates (severity, environmental exposure). Covariates are included exclusively to improve precision and reduce residual variance. Because treatment uptake may not be perfect, some treated individuals may not consistently use the kit, the analysis also estimates the Complier Average Causal Effect (CACE) using instrumental variables. Assignment Z_i serves as a valid instrument for actual usage D_i:

First stage:

$$D_i = \pi_0 + \pi_1 Z_i + \nu_i.$$

Second stage (2SLS):

$$Y_i = \alpha + \beta \widehat{D}_i + \varepsilon_i.$$

Here, β identifies the effect among compliers, and in a randomized design with monotonicity, CACE coincides with the Local Average Treatment Effect (LATE). Finally, the analysis examines heterogeneity along theoretically relevant dimensions: baseline flare frequency, socio-economic status (SES), and environmental conditions (humidity and ventilation). These subgroups reflect plausible effect modifiers and enhance the interpretability and policy relevance of the findings.

Several threats to internal and external validity need consideration. First, noncompliance may arise if some treated participants do not regularly use the kit. While this does not compromise the internal validity of the ITT estimate, since treatment assignment remains random, it attenuates the estimated treatment contrast. For this reason, CACE estimates are reported as complementary evidence to isolate effects among compliers. Second concern relates to measurement errors in outcomes such as flare days, which partly rely on self-reported diaries. These would bias estimates toward zero, reducing statistical power and potentially masking clinically relevant effects. To reduce this risk, the study combines validated clinical assessments (EASI) with short recall periods through weekly diaries, while maintaining assessor blinding to treatment status. The risk of contamination is further limited by the eligibility criteria, which exclude households already using allergen control tools, thereby reducing the likelihood that control participants independently adopt similar interventions during the study period. Finally, external validity is inherently constrained to contexts with comparable environmental conditions such as humidity, ventilation, dust mite exposure and to populations with similar socioeconomic and behavioral characteristics.

The study continues with Power Analysis, the primary endpoint is the monthly flare count, modelled as a count outcome. The reduction of one flare per month is considered clinically meaningful. However, the relative magnitude of such a change depends on baseline flare intensity. For example, a decrease of one flare represents a 50% improvement when the baseline mean is 2, but only 25% when the baseline mean is 4. This proportional difference introduces uncertainty, making it inappropriate to rely on a single baseline assumption. Instead, a scenario-based power analysis is adopted. Under conventional assumptions (power 80%, $\alpha = 0.05$), a minimum detectable effect of −1 flare per month implies sample size requirements that vary across scenarios, as reported in Table 2. To ensure adequate power under higher-variance conditions, the study adopts a conservative planning target of 141 participants per arm (Scenario C). The final sample size will be re-estimated after baseline data collection using the empirically observed mean and variance of monthly flare counts.

The study is designed to be financially feasible and operationally realistic. The total budget is estimated at 157.680€ covering the provision of standardized allergen-reduction kits, staffing, data collection, monitoring, and logistical costs (Tables 3 and 4). The study timeline spans twelve months, with baseline data collection followed by randomization and kit distribution, and outcome assessments at six and twelve months. This duration is sufficient to capture both short-term behavioral responses and sustained effects on flare frequency, while remaining compatible with typical public health program cycles.

In conclusion, this study evaluates a relative low cost and scalable environmental intervention aimed at reducing exposure to indoor triggers of atopic dermatitis. By leveraging a rigorously designed randomized controlled trial, the analysis provides credible causal evidence on the effectiveness of allergen reduction tools in decreasing flare frequency and improving patient outcomes. The results

are directly relevant for public health policy, particularly in informing preventive strategies for households facing financial or informational barriers to adoption. While not a substitute for pharmacological treatment, the intervention represents a complementary, upstream approach that may reduce symptom burden and healthcare utilization, with potential for cost effective scale up in comparable environmental settings.

Tables.

Table 1. Allergen-Reduction Kit

Component	Unit Cost (€)	Description
Anti-mite mattress & pillow covers	90,00 €	Certified allergen-impermeable covers
HEPA air purifier	140,00 €	Portable unit for bedroom use
Low-irritant detergent	25,00 €	Hypoallergenic cleaning product
Instruction leaflet & materials	5,00 €	Printing, packaging, and basic educational material
Total cost per kit	**260,00 €**	
Total cost (282 N × 260 €)	**73.320 €**	

Table 2. Power Scenarios for Monthly Flare Count Outcome. Effect size = −1 flare per month, power = 0.80, α = 0.05

Scenario	Baseline Mean (μ)	Standard Deviation (σ)	Required N per Arm	Interpretation
A	2 flares/month	1.5	36	Large proportional effect (50%) low sample required
B	3 flares/month	2.0	63	Moderate relative change (33%)
C	4 flares/month	3.0	141	Smallest proportional effect (25%) → highest sample required

Source: Personal Elaboration of https://clincalc.com/stats/samplesize.aspx

Table 3. Budget Calculation

Cost Category	Estimated Cost (€)	Details
Recruitment & pre-screening	18.000€	Staff time (research assistants) for survey administration and eligibility checks
Baseline assessment & randomization	14.000 €	Clinical assessors and study coordinators
Kit procurement & distribution	16.000 €	Logistics, transport, storage
Follow-up coordination	20.000 €	Staff time for participant tracking, reminders, and surveys
Data management & analysis	16.360 €	Data analyst and project manager
Total staff & logistics	**84.360 €**	

Table 4. Overall Total Budget

Category	Total (€)
Allergen-Reduction Kits	73.320 €
Staff and Logistics	84.360 €
Overall Total Budget	**157.680 €**

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